
Numerical Methods Sample Exam

1. Consider the problem of finding a solution of the algebraic equation $e^x - 6x - 2 = 0$ in the interval $[0, 4]$.

- (a) Compute one step of Newton's algorithm, starting from the midpoint of the interval.

Newton's method: $p_{n+1} = p_n - \frac{f(p_n)}{f'(p_n)}$ with $f'(x) = e^x - 6$.

$$p_1 = p_0 - \frac{f(p_0)}{f'(p_0)}$$

$$p_1 = 2 - \frac{f(2)}{f'(2)}$$

$$p_1 = 2 - (e^2 - 12 - 2)/(e^2 - 6)$$

$$p_1 = 2 - (-6.6109)/(1.3891)$$

$$p_1 = 6.7593 \text{ (4dp)}$$

- (b) Is the value obtained within the interval $[0, 4]$? Why?

We leave the interval $[0, 4]$. Newton's algorithm doesn't have the bracket preserving property. Divergence can occur if the initial guess is far from the actual root.

- (c) Perform one step of the bisection method to find a smaller bracket for the solution. $f(0) = -1.00$, $f(4) = 28.60$ different signs so we can use bisection. $f(2) = -6.61 < 0$, new bracket should be $[2, 4]$ to ensure it contains a root.
- (d) Perform one step of Newton's method starting from the midpoint of the interval (bracket) you found in part (c).

$$p_1 = p_0 - \frac{f(p_0)}{f'(p_0)}$$

$$p_1 = 3 - \frac{f(3)}{f'(3)}$$

$$p_1 = 3 - (e^3 - 18 - 2)/(e^3 - 6)$$

$$p_1 = 2.9939 \text{ (4dp)}$$

- (e) Do you expect your new approximation to be within 0.1 of the true solution? Give a reason for your answer.

Based on the error estimate $|p_1 - p| \lesssim |p_1 - p_0| = |2.9939 - 3.0000| = 6.1e - 3$

2. Consider the problem of finding the value of $y(3)$ for the initial value problem

$$\frac{dy}{dt} = \frac{1}{t-1} - y^2, \quad y(2) = 1.4.$$

- (a) Solve the problem using Ralston's method with a step size of $h = 0.5$.

$$\omega_{i+1} = \omega_i + \frac{1}{4}h \left(f(t_i, \omega_i) + 3f\left(t_i + \frac{2}{3}h, \omega_i + \frac{2}{3}hf(t_i, \omega_i)\right) \right)$$

$$t_0 = 2, \omega_0 = y(2) = 1.4$$

$$f(t_0, \omega_0) = f(2.0000, 1.4000) = -0.9600$$

$$f\left(t_0 + \frac{2}{3}h, \omega_0 + \frac{2}{3}hf(t_0, \omega_0)\right) = f(2.3333, 1.0800) = -0.4164$$

$$\omega_1 = 1.4000 + \frac{1}{4} \times 0.5 \times (-0.9500 + 3 \times -0.4164) = 1.1239$$

Same thing for the next step $t_1 = 2.5$ and $\omega_1 = 1.1239$

$$f(t_1, \omega_1) = f(2.5000, 1.1239) = -0.5964$$

$$f\left(t_1 + \frac{2}{3}h, \omega_1 + \frac{2}{3}hf(t_1, \omega_1)\right) = f(2.8333, 0.9251) = -0.3103$$

$$\omega_2 = 1.1239 + \frac{1}{4} \times 0.5 \times (-0.5964 + 3 \times -0.3103) = 0.9329$$

- (b) How would you expect the error to change if you were to take a stepsize of $h = 0.25$?
Ralston's method is second order so the error is expected to decrease by a factor of $(0.25/0.5)^2 = (1/4)$.

3. Consider the following data

x_i	2.0	1.5	1.0	3.0
$y_i = f(x_i)$	0.40	1.00	1.82	0.03

- (a) Compute the divided differences $f[x_i, \dots, x_j]$ for $0 \leq i \leq j \leq 3$. You may assume $f[x_0, x_1] = -1.200$ and $f[x_1, x_2] = -1.640$.
- (b) Write down the nested form of the cubic polynomial p_3 interpolating the given data.
- (c) Estimate the value of y when $x = 2.5$, showing intermediate steps in your computations.

Divided differences

$$f[x_i, \dots, x_j] = \frac{f[x_{i+1}, \dots, x_j] - f[x_i, \dots, x_{j-1}]}{x_j - x_i}$$

$$f[x_2, x_3] = \frac{0.03 - 1.82}{3.0 - 1.0} = -0.8950$$

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = 0.4400$$

$$f[x_1, x_2, x_3] = \frac{f[x_2, x_3] - f[x_1, x_2]}{x_3 - x_1} = 0.4967$$

$$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0} = 0.0567$$

The cubic polynomial in nested form is

$$p_3(x) = 0.40 + (x - 2.0)(-1.200 + (x - 1.5)(0.4400 + (x - 1.0)0.0567))$$

The value of y when $x = 2.5$ is $p(2.5) = 0.0625$ (4dp) and $p(2.5) = 0.06$ (2dp)

4. Consider the problem of computing to an accuracy of 0.01 the integral

$$I = \int_{0.0}^{1.2} \cos(x^2) dx.$$

- (a) Use the trapezoid rule with $n = 4$ subintervals to estimate the value of I . Work in 4dp.

Stepsize $h = 1.2/4 = 0.3$, x_i are $(0.0, 0.3, 0.6, 0.9, 1.2)$.

$$T_4 = h \left(\frac{1}{2}f(x_0) + \sum_{i=1}^3 f(x_i) + \frac{1}{2}f(x_4) \right)$$

$$T_4 = 0.3 \times \left(\frac{1}{2} \times 1.0000 + 0.9960 + 0.9359 + 0.6895 + \frac{1}{2} \times 0.1304 \right)$$

$$T_4 = 0.9560$$

- (b) Use the error estimate for the trapezoid rule to estimate the error over the subintervals $[0.0, 0.6]$ and $[0.6, 1.2]$. Do you expect the error to be less than the desired accuracy bound?

Over $[0.0, 0.6]$, the error estimate is

$$\frac{1}{3}|T_2(f; [0.0, 0.6]) - T_1(f; [0.0, 0.6])| = |0.5892 - 0.5808| = 2.8 \times 10^{-3}$$

Over $[0.6, 1.2]$, the error estimate is

$$\frac{1}{3}|T_2(f; [0.6, 1.2]) - T_1(f; [0.6, 1.2])| = |0.3668 - 0.3199| = 1.6 \times 10^{-2}$$

Total error is $2.8 \times 10^{-3} + 1.6 \times 10^{-2} \approx 1.8 \times 10^{-2}$, so the error is not expected to be less than 0.01.

- (c) Explain how to use the adaptive trapezoid method to obtain a better estimate to the integral. (You do not have to perform the computation).
Iteratively subdivide one of the intervals to reduce the expected error on that interval (about 1/4). At each iteration pick the interval with the largest estimate on the error.

5. The least squares approximation to a function f over the interval $[-1, 1]$ is given by

$$q_n(x) = \sum_{k=0}^n c_k P_k(x)$$

where P_k are the Legendre polynomials. Suppose c_k for $k = 0, \dots, 5$ are given by

k	0	1	2	3	4	5
c_k	0.3086	0.2193	-0.0842	-0.1233	-0.0267	0.0262

- (a) Use the recurrence equation to calculate $P_k(0.7)$ for $k = 0, 1, \dots, 4$.

$$P_0(0.7) = 1.0000$$

$$P_1(0.7) = x = 0.7000$$

$$P_2(0.7) = \left(2 - \frac{1}{2}\right) \times 0.7 \times P_1(0.7) - \left(1 - \frac{1}{2}\right) \times P_0(0.7) = 0.2350$$

$$P_3(0.7) = \left(2 - \frac{1}{3}\right) \times 0.7 \times P_2(0.7) - \left(1 - \frac{1}{3}\right) \times P_1(0.7) = -0.1925$$

$$P_4(0.7) = \left(2 - \frac{1}{4}\right) \times 0.7 \times P_3(0.7) - \left(1 - \frac{1}{4}\right) \times P_2(0.7) = -0.4121.$$

- (b) Compute $q_4(0.7)$.

$$q_4(0.7) = \sum_{k=0}^4 c_k P_k(0.7) = 0.4771$$

- (c) Explain how orthogonality of the P_k gives rise to the following simple formula for the square integral

$$\int_{-1}^1 q_n(x)^2 dx = \sum_{k=0}^n c_k^2 \int_{-1}^1 P_k(x)^2 dx.$$

We can write the integral as

$$\sum_{i,j} c_i c_j \int_{-1}^1 P_i(x) P_j(x) dx$$

Because $\int_{-1}^1 P_i(x) P_j(x) dx = 0$ when $i \neq j$ by the orthogonality of Legendre polynomials, we are left only with the terms $\int_{-1}^1 c_i^2 P_i(x)^2 dx$.

(d) Compute the square integral $\int_{-1}^1 q_2(x)^2 dx$.

$$\int_{-1}^1 q_2(x)^2 dx = \sum_{k=0}^2 \alpha_k c_k^2.$$

$$\alpha_k = 1/(k + \frac{1}{2}), \text{ and } \int_{-1}^1 q_2(x)^2 dx = 0.2254.$$

6. Let

$$A = A^{(0)} = \begin{pmatrix} 5 & 2 \\ 2 & 3 \end{pmatrix}.$$

After applying one step of QR method we obtain

$$A^{(0)} = Q^{(0)}R^{(0)} = \begin{pmatrix} 0.9285 & -0.3714 \\ 0.3714 & 0.9285 \end{pmatrix} \begin{pmatrix} 5.3852 & 2.9711 \\ 0 & 2.0426 \end{pmatrix}; \quad A^{(1)} = R^{(0)}Q^{(0)} = \begin{pmatrix} 6.1036 & 0.7586 \\ 0.7586 & 1.8966 \end{pmatrix}.$$

(a) Apply one step of the QR method to the matrix $A^{(1)}$.

$$\begin{aligned} v_1 &= x_1 = (6.1036, 0.7586)^T \\ v_2 &= x_2 = \frac{x_2 \cdot v_1}{v_1 \cdot v_1} v_1 v_2 = (-0.2206, 1.7748)^T \\ r_{11} &= \|v_1\| = 6.1504 \\ r_{22} &= \|v_2\| = 1.7885 \\ u_1 &= \frac{v_1}{\|v_1\|}, \quad u_2 = \frac{v_2}{\|v_2\|} \\ u_1 &= (0.9924, 0.1233)^T, \quad u_2 = (-0.1233, 0.9924)^T \\ r_{12} &= x_2 \cdot u_1 = 0.9868, \quad r_{21} = x_1 \cdot u_2 = 0.0000 \\ A^{(2)} &= R^{(1)}Q^{(1)} = \begin{pmatrix} 6.2252 & 0.2206 \\ 0.2206 & 1.7748 \end{pmatrix} \end{aligned}$$

(b) Use your answer to part (a) to estimate the eigenvalues of A.

Eigenvalue estimates are the diagonal entries of A. $\lambda_1 \approx 6.2252$, $\lambda_2 = 1.7748$